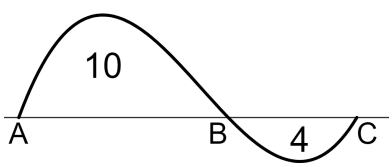
# Review of Antiderivatives

First, you need to make sure you refresh yourself on the antiderivatives below, since they are the foundation of ALL integrals!! I will refer to these as "Day 1 Integrals" throughout the semester.

Table of Some Antiderivatives (Day 1 Integrals):

| Function | General Antiderivative | Function | General Antiderivative | Function | General Antiderivative |
|---|---|---|---|---|---|
| $a$(a constant) | $ax + C$ | $\frac{1}{\sqrt{1-x^2}}$ | $\arcsin x + C$ | $\sec^2 x$ | $\tan x + C$ |
| $x^n, n \neq -1$ | $\frac{x^{n+1}}{n+1} + C$ | $\frac{1}{1+x^2}$ | $\arctan x + C$ | $\sec x \tan x$ | $\sec x + C$ |
| $\frac{1}{x}$ | $\ln\lvert x\rvert + C$ | $\sin x$ | $-\cos x + C$ | $\csc x \cot x$ | $-\csc x + C$ |
| $e^x$ | $e^x + C$ | $\cos x$ | $\sin x + C$ | $\csc^2 x$ | $-\cot x + C$ |
| $a^x$ | $\frac{a^x}{\ln a} + C$ | | | | |

The **definite integral** $\int_a^b f(x)\ dx$ represents the NET AREA bounded between $f(x)$ and the $x-$axis on the interval $[a, b]$. Suppose the area of each region between the curve $f(x)$ and the $x-$axis is shown in the figure.




$\int_A^B f(x)\ dx =$ 10

$\int_B^C f(x)\ dx =$ -4

$\int_A^C f(x)\ dx =$ 6

What is the total area bounded between the curve and the $x-$axis? 14

To evaluate definite integrals, we use the following:

The Fundamental Theorem of Calculus, Part 2:

If $f$ is continuous on $[a, b]$, then

$$\int_a^b f(x)\ dx = F(b) - F(a)$$

where $F$ is any antiderivative of $f$.

## 5.5: Substitution

Recall the Chain Rule for derivatives: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$

$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$ $\quad \frac{d}{dx}\sin(g(x)) = \cos(g(x))g'(x)$ $\quad \frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x)$ $\quad \frac{d}{dx}\ln(g(x)) = \frac{g'(x)}{g(x)}$

The **Substitution Rule** for integrals is essentially the reverse of the Chain Rule: If $u = g(x)$ is a differentiable function, then

$$\int f(g(x))g'(x)\ dx = \int f(u)\ du$$

1. (a) Find $f'(x)$ if $f(x) = \left(x^2+1\right)^{100}$

   (b) Evaluate $\int 100\left(x^2+1\right)^{99} \cdot 2x\ dx$

   (c) What if we didn't have part (a)?

2. $\int te^{1-4t^2}\ dt$

3. $\int \frac{50e^{5z}+100}{e^{5z}+10z}\,dz$

$u = e^{5z} + 10z$

$du = 5e^{5z} + 10\,dz$

$du = 5(e^{5z}+2)\,dz$

$\frac{1}{5(e^{5z}+2)}\,du = dz$

$\int \frac{50(e^{5z}+2)}{e^{5z}+10z}\,dz$

$\frac{50}{5}\int u\,du$

$10\ln|e^{5z}+10z| + C$

4. $\int \frac{1}{x\ln(x)}\,dx$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$\int \frac{1}{u}\,du = \ln|u| \Rightarrow \ln(\ln x)$

5. $\int \frac{\arcsin 4x}{\sqrt{1-16x^2}}\,dx$

$u = \sin^{-1}(4x)$

$du = \frac{1}{\sqrt{1-16x^2}} \cdot 4\,dx$

$\frac{1}{4}du = \frac{1}{\sqrt{1-16x^2}}\,dx$

$\int \arcsin(4x) \cdot \frac{1}{\sqrt{1-16x^2}}\,dx$

$\frac{1}{4}\int u\,du \Rightarrow \frac{1}{4}\frac{u^2}{2}$

$= \frac{1}{8}(\sin^{-1}(4x))^2 + C$

$\rightarrow$ let u be sin x

6. $\int \tan x \, dx = \int \frac{\sin x}{\cos x}$

$u = \cos x$

$du = -\sin x \, dx$

$-\int \frac{1}{u} du$

$= -\ln|\cos x| + C$

7. (a) $\int e^{3x} \, dx = \frac{1}{3} e^{3x} +$

(b) $\int \sin(11x - 5) \, dx$

$u = 11x - 5$

$du = 11 \, dx$

$\frac{1}{11} \int \sin u \, du$

$= -\frac{1}{11} \cos(11x - 5) + C$

(c) $\int \sec^2(x - 8) \, dx = \tan(x - 8) + C$

8. $\int \frac{5+7x}{x^2+1}\,dx$

$u = x^2+1$

$du = 2x\,dx$

$$\int \frac{5}{x^2+1} + \int \frac{7x}{x^2+1}$$

$$5\tan^{-1}(x) + \frac{7}{2}\int \frac{1}{u}\,du$$

$$5\tan^{-1}(x) + \frac{7}{2}\ln|x^2+1| + C$$

9. (a) $\int x(x+9)^7\,dx$

$u = x+9 \qquad x = u-9$

$du = dx$

$$\int (u-9)\,u^7\,du$$

$$\int u^8 - 9u^7\,du$$

$$= \frac{u^9}{9} - \frac{9u^8}{8} \Rightarrow \frac{(x+9)^9}{9} - \frac{9}{8}(x+9)^8$$

(b) $\int \frac{x^3}{\sqrt{5x^2-1}}\,dx$

$u = 5x^2-1 \rightarrow \frac{u+1}{5}$

$du = 10x\,dx$

$$= \int x^2 \cdot \frac{x}{\sqrt{5x^2-1}}\,dx$$

$$= \frac{1}{10}\int \frac{u+1}{5}\cdot\frac{1}{\sqrt{u}}\,du = \frac{1}{50}\int \frac{u+1}{\sqrt{u}} = \frac{1}{50}\left(\int \frac{u}{\sqrt{u}} + \int \frac{1}{\sqrt{u}}\right)$$

$$= \frac{1}{50}\left(\int u^{1/2} + u^{-1/2}\right)du \Rightarrow \frac{1}{50}\left(\frac{2}{3}u^{3/2} + 2u^{1/2}\right)$$

$$= \frac{1}{75}(5x^2-1)^{3/2} + \frac{1}{25}(5x^2-1)^{1/2} + C$$

For *definite integrals*, there are two ways to evaluate when using substitution:

10. (a) $\displaystyle\int_4^9 \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$

$u = \sqrt{x}$

$du = \frac{1}{2}\frac{1}{\sqrt{x}}\,dx$

$2du = \frac{1}{\sqrt{x}}\,dx$

$2\displaystyle\int_{x=4}^{x=9} e^u\,du$

$2\,e^u\Big|_{x=4}^{x=9}$

$2\left(e^{\sqrt{x}}\right)\Big|_{x=4}^{x=9} \Rightarrow 2\left(e^3 - e^2\right)$

(b) $\displaystyle\int_4^9 \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$

$u = \sqrt{x}$

$du = \frac{1}{2}\frac{1}{\sqrt{x}}\,dx$

$2du = \frac{1}{\sqrt{x}}\,dx$

$2\displaystyle\int_2^3 e^u\,du$

$= 2\left(e^3 - e^2\right)$

11. Use substitution to rewrite the integral as a Day 1 Integral (really two Day 1 Integrals):

$\displaystyle\int_1^4 \frac{x}{6-x}\,dx$

$u = 6 - x \Rightarrow x = 6 - u$

$du = -dx$

$-du = dx$

Bounds: $u(4) = 2$

$u(1) = 5$

$-\displaystyle\int_5^2 \frac{6-u}{u}\,du$

$\Rightarrow \displaystyle\int_2^5 \frac{6-u}{u} = \int_2^5 \left(\frac{6}{u} - 1\right)$

12. If $f$ is differentiable and $\displaystyle\int_{1/4}^1 f(u)\,du = 12$, find $\displaystyle\int_1^2 \frac{f(1/x^2)}{x^3}\,dx$.

$u = \frac{1}{x^2} = x^{-2}$

$du = -2\frac{1}{x^3}\,dx$

Bounds: $u(2) = \frac{1}{4}$

$u(1) = 1$

$-\frac{1}{2}\displaystyle\int_1^{1/4} f(u)\,du$

$= \frac{1}{2}\underbrace{\displaystyle\int_{1/4}^1 f(u)\,du}_{12}$

$\Rightarrow \frac{1}{2}(12) = 6$